SETS GAMEATHON 2026 - FINAL ROUND SUBMISSION

PROTOTYPE SOLUTION SUMMARY: CYBERQUEST RPG

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1. PROBLEM STATEMENT

Today, many users fall victim to social engineering and malicious attacks due to a critical lack of cybersecurity awareness. Our digital environment has grown increasingly complex, exposing ordinary citizens to sophisticated threat vectors such as phishing, malware update prompts, and AI-based cybercrimes like voice-cloned vishing calls.

Key Vulnerability Challenges:

- **Phishing Attempts & Domain Deception:** Users lack the reflexes to scrutinize domain links, often clicking on deceptive homograph URLs containing character lookalikes (e.g. Cyrillic or non-standard characters).
- **High-Pressure Social Engineering:** Vishing callers manipulate human emotions (fear, panic, urgency). Traditional text-based security protocols fail because they do not train users under simulated pressure.
- **Malware updates & Mirrors:** Malicious browser update notifications trick non-technical users into executing scripts from unofficial repositories.
- **Traditional Training Limitations:** Static pamphlets, slide presentations, and compliance videos are flat and passive. They fail to build active, muscle-memory-based defenses.
- **Accessibility Gaps:** Most training platforms overlook users with disabilities (such as hearing impairments), failing to provide inclusive solutions.

2. SYSTEM OBJECTIVES

The primary objective is to develop an innovative, lightweight web application that boosts cybersecurity awareness and reduces users' vulnerability to attacks by replacing passive reading with active, gamified decision-making. The system achieves this through the following core targets:

- **Spot and Reject Phishing:** Train users to check link details before clicking, verifying hostnames and spotting deceptive symbols.
- **Spot Social Engineering:** Empower users to recognize pressure vishing call behaviors and deny sharing OTPs or credentials.
- **Enforce Actionable Defenses:** Familiarize users with active protocols including Multi-Factor Authentication (MFA), password complexity, and reporting scams to official channels like the National Cybercrime Portal (1930 / cybercrime.gov.in).
- **Failure Integration:** Establish a zero-risk sandbox where players experience the consequences of their mistakes, creating muscle memory to act safely in real scenarios.
- **Inclusive Accessibility:** Design an app that accommodates varied hardware capabilities and users with cognitive or hearing impairments.

3. PROTOTYPE OVERVIEW (CYBERQUEST RPG)

CyberQuest RPG is a web-based, gamified 3D training platform where players become digital realm guardians navigating corridors representing cybercrime vectors. The application combines interactive scenarios, server-side AI, and real-time feedback.

3.1 Technical Architecture:

- **Frontend:** Built using vanilla HTML, CSS, JavaScript, and Three.js (r128 WebGL). Runs directly in modern browsers on both mobile and desktop without requiring engine downloads or installations.
- **Backend Services:** Deployed on Firebase Hosting. Utilizes Cloud Firestore for leaderboard data tracking, and Anonymous Firebase Authentication to ensure player onboarding is immediate and completely private.
- **AI Key Security:** Client requests are routed through a server-side Firebase Cloud Function proxy (/api/guide). This keeps the Groq Chat Completions API key hidden, protecting it from reverse engineering in the client browser.

3.2 Core Gameplay & Simulated Threat Scenarios:

- **Phishing Homograph Maze:** Simulates fake login triggers. Players must hover over the verify link to trigger a tooltip exposing the true URL destination (such as arcadiakyc-login.org, containing a lookalike dotless Turkish i symbol). Clicking on deceptive links drains player health (HP).
- **Social Engineering Bank Simulator:** Simulates high-pressure voice scams (e.g. Arcadia Bank Fraud Team). Prompts the player with real-world pressure dialogs asking for recovery codes. The player must choose to hang up and verify, or fail the level.
- **Malware & update checks:** Deceptive alert overlays block labyrinth pathways. Players learn to identify unverified downloads and select official update mirrors.
- **Ask the Cyber Guide (AI Mentor):** Players can open a console to converse with an AI guide ('Greymantle') powered by serverless Groq completions, asking custom cybersecurity questions or purchasing gameplay clues.
- **Void Collective Checkpoint Gates:** Checkpoint gates block access keys. Players are presented with 3 sequential, non-technical cybersecurity multiple-choice questions. They must answer them sequentially and score at least 2/3 correct to pass the gate.

3.3 Active Score Performance Formula:

To evaluate behavioral success, scores are calculated dynamically using the formula: Score = XP + (Accuracy% x 2) + HP. This rewards correct threat recognition (XP), precise decision accuracy, and survival metrics (HP remaining). Result records are logged anonymously on the Firestore database.

3.4 Inclusivity & Accessibility Support:

- **Hearing Impaired Subtitles:** Simulated audio vishing calls feature immediate on-screen textual subtitles and transcripts, allowing deaf or hearing-impaired users to complete the training seamlessly.
- **Physical Controls Adaptation:** Features simplified keyboard mappings (WASD/Arrows + Enter keys) for all navigation and dialog checkpoints, supporting players with motor limitations.
- **Visual Adjustments:** High-contrast dark theme (#0B0E14), responsive sizing, clear warnings (red highlight traps) and gold indicators help accommodate users with color blindness and cognitive load variations.
- **Hardware Constraints:** Highly compressed client bundle. Executes cleanly on integrated graphics cards and standard browsers without requiring downloads, rendering in under 1.5s on mobile or low-spec systems.

Submission link: https://cyberquest-rpg.web.app